

Young–Laplace initial conditions for bursting-bubble simulations

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Abstract

A bursting-bubble computation starts from an open cavity at a free surface. That cavity is the Young–Laplace equilibrium of a prescribed gas volume. The Bond number based on the equivalent-sphere radius sets how far gravity pulls the bubble away from a sphere. We document the Python generator that builds these shapes for Basilisk Stage 1. Bond continuation from a cheap spherical seed replaces the old one-shot curvature guess. At vanishing Bond the meniscus match is undefined, so the generator writes a unit sphere meeting a plane, regularized by a fillet of scale δ . The opening angle recovers the comparison of Lhuissier and Villiermaux [2012], and the written polyline sits on the repository’s shipped $\mathcal{Bo} = 10^{-3}$ file.

1 Introduction

When a bursting-bubble computation starts, the thin film that closed the cavity has already gone. The remaining geometry is an open cavity meeting a free surface. That shape is not a drawing choice. It is the quasi-static Young–Laplace equilibrium of a gas volume at a liquid–air interface [Princen, 1963], and it sets the first capillary waves that later focus at the bottom of the cavity.

The Bond number

$$\mathcal{Bo} = \frac{\rho g R_0^2}{\gamma} \quad (1)$$

compares gravity to surface tension on the equivalent-sphere radius R_0 , the radius of the sphere with the same gas volume. Lengths in this report, and in the Basilisk solvers, are scaled by R_0 . That choice is not the same as a Bond number based on the bottom curvature radius. At small Bond the cavity is nearly spherical. As Bond increases the bubble flattens, the submerged profile lengthens, and the opening left by the ruptured film widens. Lhuissier and Villiermaux [2012] mapped that opening against the capillary length. Direct numerical studies of bursting, including our viscoplastic calculations [Sanjay et al., 2021] and the Newtonian cross-validation of Deike et al. [2018], have typically taken a near-spherical cavity at $\mathcal{Bo} = 10^{-3}$ rather than solving the equilibrium at the Bond number of the run.

The MATLAB driver `InitialCondition.m` that accompanies Lhuissier and Villiermaux [2012] already solves this nested shooting problem. Working copies of that driver each carried a different first guess for the bottom curvature. A cold start at large Bond with a guessed $R_b^{\max}$ is fragile. The Python generator in `simulationCases/initialConditions/` ports that construction, walks Bond upward from 10^{-3} , and writes the polyline that Stage 1 of the Basilisk bursting-bubble solvers [Popinet, 2009] consumes. At strictly vanishing Bond the $2/\mathcal{Bo}$ meniscus match is undefined. That limit is a unit sphere meeting a plane, regularized by a fillet of scale δ .

The report is organized as follows. §2 states the equilibrium problem. §3 describes the nested shoot and the continuation. §4 treats the neck fillet. §5 is the Bond-zero geometry. §6 records the file convention and the checks against Lhuissier and Villiermaux [2012]. §7 is the command line. We close with the limitations of a quasi-static initial cavity.

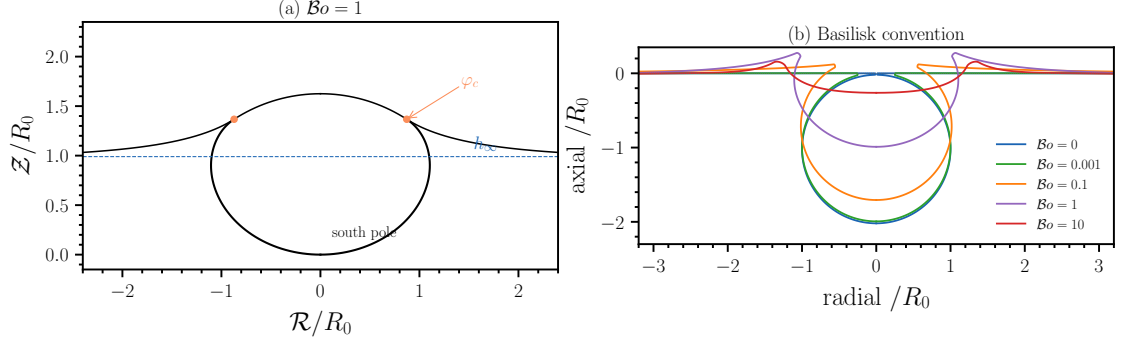


Figure 1: Equilibrium construction. (a) Submerged cavity, spherical cap, and outer meniscus at $Bo = 1$, plotted with $\mathcal{Z}$ upward from the south pole. The contact station is φ_c ; the opening is $\alpha_c = \pi - \varphi_c$. (b) The same generator written in the Basilisk convention, cavity in $-$ axial, free surface at axial = 0, for $Bo = 0, 10^{-3}, 0.1, 1$, and 10 . The $Bo = 0$ curve is the sphere-plane construction of §5.

2 Equilibrium problem

The generating curve is axisymmetric. The tangent angle φ is measured from the horizontal, starting at the south pole ($\varphi = 0$) and running toward the top ($\varphi = \pi$). Three pieces meet at a contact station φ_c .

The submerged cavity $(\mathcal{R}(\varphi), \mathcal{Z}(\varphi))$ satisfies the Young–Laplace balance written as an ODE in φ ,

$$\frac{d\mathcal{R}}{d\varphi} = \frac{\mathcal{R} \cos \varphi}{\mathcal{R} (2/R_b + Bo \mathcal{Z}) - \sin \varphi}, \quad (2)$$

$$\frac{d\mathcal{Z}}{d\varphi} = \frac{\mathcal{R} \sin \varphi}{\mathcal{R} (2/R_b + Bo \mathcal{Z}) - \sin \varphi}. \quad (3)$$

The south pole is regularized by a small start $\varphi_0 = 10^{-5}$ with $(\mathcal{R}, \mathcal{Z}) = (R_b \sin \varphi_0, R_b(1 - \cos \varphi_0))$. The bottom curvature radius R_b is unknown. We use the $\sin \varphi$ form of (3) rather than $\tan \varphi d\mathcal{R}/d\varphi$, so the equator stays regular.

A spherical cap of radius R_c matches slope and curvature at the contact. The far-field free surface is an outer meniscus that must approach a constant height h_∞ as $\mathcal{R} \rightarrow \infty$. Pressure balance between the south pole, the cap, and that far field fixes

$$h_\infty = \frac{2}{Bo} \left(\frac{2}{R_c} - \frac{1}{R_b} \right). \quad (4)$$

The enclosed gas volume, scaled by $4\pi/3$, is

$$\frac{3}{4} \int \mathcal{R}^2 d\mathcal{Z} = 1, \quad (5)$$

with the integral taken over the submerged profile and the spherical cap.

The opening angle of Lhuissier and Villiermaux [2012] is $\alpha_c = \pi - \varphi_c$. Their small-Bond estimate, when $R_c \sim R_0$, is

$$\alpha_c = \frac{1}{2\sqrt{3}} \frac{R_c}{a}, \quad a = \sqrt{\gamma/\rho g} = R_0/\sqrt{Bo}. \quad (6)$$

Figure 1 shows one hit at $Bo = 1$ in the integration coordinates, together with the family of written Basilisk polyines.

3 Nested shooting and Bond continuation

Two unknowns remain, R_b and φ_c . The outer loop shoots R_b so that (5) holds. For each trial R_b the inner loop shoots φ_c so that the meniscus meets h_∞ .

The meniscus residual is not $Z(R_{\max}) - h_\infty$. Beyond a few capillary lengths the decaying meniscus mode is swamped by the growing one, and that far-field difference jumps discontinuously with φ_c . We match at $\mathcal{R}_c + O(1/\sqrt{\mathcal{B}o})$, a station a few capillary lengths out from the contact. The MATLAB driver integrated to a fixed `TailxMax` of 8 (sometimes 4) and only afterwards padded the free surface to 32. Checking at infinity is neither necessary nor stable.

Linear interpolation of $(\mathcal{R}, \mathcal{Z})$ onto φ_c uses the local interval $\varphi_{i+1} - \varphi_i$. The MATLAB driver used a broken denominator here. That bug is silent at small Bond, where φ_c sits near π and the samples are dense. It is not silent once the contact walks down the profile.

A volume residual that does not change sign is not reported as a root. The R_b bracket expands, first from a previous hit if one exists, and otherwise from a Bond-dependent upper bound. Failures return an error rather than a guessed polyline.

Continuation replaces the old per-Bond R_b guesses. Small-Bond shapes are nearly spherical ($R_b \approx 1$). Walking $\mathcal{B}o$ upward, using each accepted (R_b, φ_c) as the next guess, is much more stable than a cold start at large Bond. The ladder starts at 10^{-3} and grows by a ratio of at most 2 until the target. A comma-separated Bond list continues from the previous requested value. A prior shape is reused only when its Bond is at or below the target. Reusing a higher-Bond hit for a lower target writes the wrong geometry into `Bo%5.4f.dat`.

The Young–Laplace shooter refuses $\mathcal{B}o \leq 0$ and $\mathcal{B}o < 10^{-6}$. Those limits belong to §5.

4 Neck fillet

Removing the film leaves a curvature singularity where the cavity meets the free surface. Stage 1 cannot ingest that corner. The MATLAB driver inserted a circular fillet whose radius can change sign with the sample index. We replace that construction with a C^1 cubic Hermite blend on the contact-truncated profiles $(\mathcal{R}_{\text{sub}}, \mathcal{Z}_{\text{sub}}, \varphi_{\text{sub}})$ and the nearby meniscus. Position and tangent are matched at both cuts. The default physical span along the meniscus is $0.22 R_0$.

The blend must start at the neck. Walking from the south pole and cutting after a fixed arclength lands the fillet on the apex, and the written polyline then returns to the axis after leaving it. The generator cuts from the contact, not from the pole.

The fillet is a numerical regularisation. It is not a physical film thickness, and the subsequent bursting dynamics must be independent of its finite curvature once that curvature is high enough [Sanjay et al., 2021].

5 Vanishing Bond

Equation (4) has no $\mathcal{B}o = 0$ limit. The outer meniscus becomes a flat plane, the submerged profile becomes a unit sphere, and the contact is a point. A point contact is not a meshable initial hole.

The regularized geometry is a unit circle meeting a line, run as axisymmetric (figure 2). A fillet of scale δ opens a neck of radius $\sim 2\delta$. The construction follows the circle–line helpers in `comphy-lab/Circle-Contacts-Line`. With the sphere centre at $-(1 + \delta)$ and

$$\varphi_{c,1} = \arcsin(2\delta), \quad (7)$$

the fillet radius is

$$R_f = \frac{1 + \delta}{\cos \varphi_{c,1}} - 1. \quad (8)$$

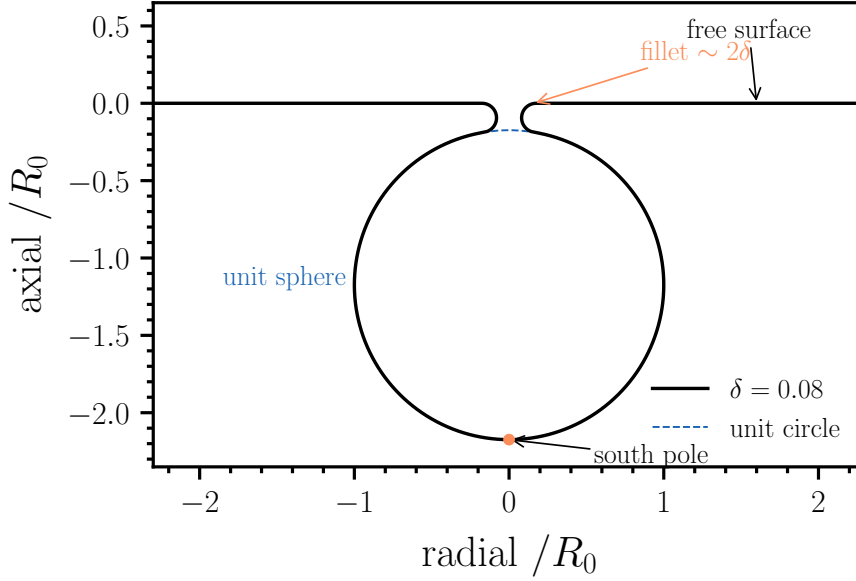


Figure 2: Regularized $\mathcal{B}o = 0$ geometry. A unit sphere meets a plane. The fillet of scale δ opens a neck of radius $\sim 2\delta$ and replaces the point contact. The schematic uses $\delta = 0.08$ so the neck is visible. The generator default is $\delta = 0.01$. Coordinates are the Basilisk convention of §6.

The curve is then shifted so the neck sits at axial = 0, mirrored into the cavity, and padded with a flat free surface out to a requested $R_{\max}$. The south pole sits near -2 , as in the finite-Bond files.

The default $\delta = 0.01$ is the value used by the drill-resolution solvers for the initial hole. The generator rejects non-finite δ or $R_{\max}$, $\delta \geq 1/2$ (where $\arcsin(2\delta)$ ceases to make sense), and a sample count below 2.

6 File convention and checks

Basilisk Stage 1 reads

```
simulationCases/DataFiles/Bo%5.4f.dat
```

through `distance.h` and `input_xy`. Column 1 is axial. Column 2 is radial. The cavity lies in $-\text{axial}$. The far-field free surface sits at axial = 0. The south pole is the first sample. `distance.h` is MPI-incompatible, so this polyline is converted to a restart dump in serial and then reused.

The repository ships `Bo0.0010.dat` as the current default Stage 1 cavity. Regenerating that Bond number from a cold start places the south pole within $0.01 R_0$ of the shipped file, keeps the volume residual below 10^{-6} , and recovers the small-Bond opening (6) to better than eight percent. A cold start at $\mathcal{B}o = 1$ walks the continuation ladder and does not return to the axis after leaving the south pole.

Figure 3 rebuilds the historical comparison. The abscissa is $(R_c/R_0)\sqrt{\mathcal{B}o} = R_c/a$. The ordinate is $2\alpha_c/\pi$. The present generator tracks the digitised points of Lhuissier and Villiermaux [2012] from $\mathcal{B}o = 10^{-4}$ to $\mathcal{B}o = 900$, including the small-Bond line (6) and the large-Bond saturation $2\alpha_c/\pi \rightarrow 1$. That comparison is a literature check of the shooter. It is not a claim about any particular bursting campaign.

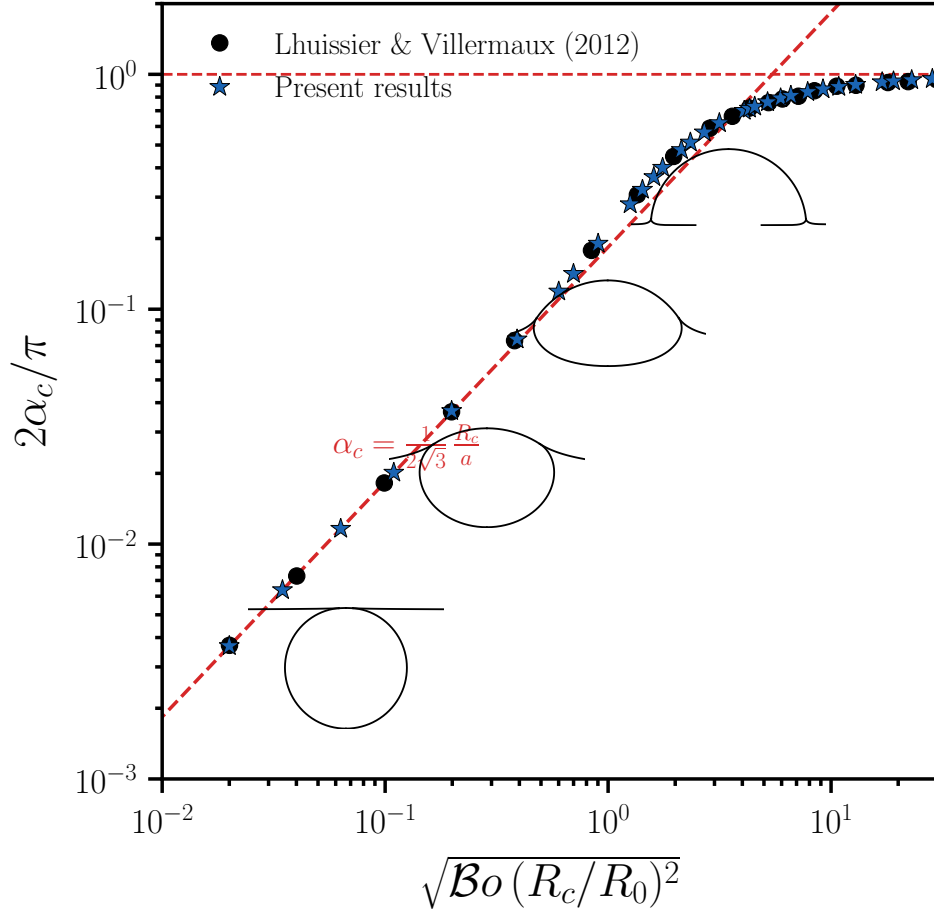


Figure 3: Opening angle against the capillary length. Filled circles are the digitised data of Lhuissier and Villiermaux [2012]. Stars are the present continuation. The dashed line at small R_c/a is (6). The upper dashed line is $2\alpha_c/\pi = 1$. Insets show the equilibrium cavity at selected Bond numbers.

7 Generating shapes

The finite-Bond driver is `generate_bond_shape.py`. The Bond-zero driver is `generate_zero_bond.py`. Both write the Stage 1 convention of §6.

```
./venv/bin/python generate_bond_shape.py --bond 0.001
./venv/bin/python generate_bond_shape.py \
    --bond 0.01,1,10 --out-dir ../DataFiles
./venv/bin/python generate_zero_bond.py --delta 0.01 \
    --out ../DataFiles/Bo0.0000.dat
```

Pass `--no-continue` to force a one-shot solve at every Bond in a list. That flag is a robustness check. It is not the recommended path at large Bond. Geometry tests live in `test_shapes.py`. The opening-angle figure is rebuilt by `plot_opening_angle.py`.

8 Concluding remarks

The initial cavity of a bursting-bubble computation is a Young–Laplace problem, not a sphere with a hole cut in it. Once Bond is finite the nested shoot on R_b and φ_c , continued from a spherical seed, gives a polyline that Stage 1 can ingest and that recovers the opening angle of Lhuissier and Villiermaux [2012]. At vanishing Bond the same file convention is filled by a regularized sphere–plane geometry.

We caution that this is a quasi-static construction after the film has gone. The true shape of a bubble that has risen to a free surface depends on how it got there, and on the rheology of the pool [Sanjay et al., 2021]. The fillet regularizes a singularity. It does not restore the missing film. Those caveats belong to the bursting calculation that consumes the file, not to the equilibrium generator.

References

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